Breast Cancer Risk Modeling and Clinical Decision Support Using Interpretable Machine Learning and Synthetic Survival Cohorts

1 Abstract

This project addresses breast cancer risk prediction and survival analysis using structured clinical data from the METABRIC dataset. The methodology combines supervised machine learning models for risk classification and survival analysis techniques to model time-to-event outcomes. The overall pipeline includes data preprocessing and feature engineering, classification models for risk stratification, survival modeling using Cox Proportional Hazards, and model evaluation with interpretability via SHAP. Three model architectures are developed: a binary benign/malignant classifier (Logistic Regression and Random Forest), a multi-class risk tier stratification model (XGBoost), and a survival analysis model (Cox Proportional Hazards). Results demonstrate that combining classification and survival approaches provides a more comprehensive clinical decision support framework than any single model alone.

2 Introduction

Breast cancer screening and diagnosis rely on clinical measurements and imaging-derived features to determine whether a tumor is likely benign or malignant. Even when a classification model is accurate, it is often not sufficient for real-world use because clinicians also need risk-level stratification, confidence they can trust, and interpretable reasons behind a prediction.

The purpose of this methodology is to:

(i) Predict patient risk levels (e.g., high vs. low risk);

(ii) Estimate survival probabilities over time;

(iii) Identify clinically significant features influencing outcomes.

Using both classification and survival models is important because classification provides discrete risk categories, whereas survival analysis captures time-to-event dynamics that are critical in clinical settings. Comparing multiple models allows us to evaluate predictive performance across approaches, understand trade-offs between interpretability and accuracy, and select models that are both clinically meaningful and statistically robust.

3 Problem Statement

This project addresses the challenge of building a clinically usable breast cancer prediction system using structured tumor features. Specifically, the project builds models that:

(i) Predict whether a tumor is malignant vs. benign;

(ii) Stratify patients into Low, Medium, and High risk tiers for decision support;

(iii) Provide confidence-calibrated probabilities;

(iv) Generate and use synthetic time-to-event survival cohorts to model disease progression risk (time-to-event and censoring) when real survival labels are not available in the chosen dataset.

4 Data Collection and Preparation

4.1 Data Sources

Two primary data sources are employed in this study.

METABRIC Dataset. The breast cancer METABRIC (Molecular Taxonomy of Breast Cancer International Consortium) database is publicly available on the cBioPortal for Cancer Genomics. A CSV download was retrieved from Kaggle.[1] The dataset and associated code are maintained in a public GitHub repository.[2]

Wisconsin Diagnostic Breast Cancer (WDBC) Dataset. A second breast cancer dataset from the Wisconsin Diagnostic Database was used for binary classification of benign and malignant tumors, retrieved from the UC Irvine Machine Learning Repository.[3]

[1] https://www.kaggle.com/datasets/gunesevitan/breast-cancer-metabric
[2] https://github.com/anudeshpande24/Capstone_Cancer_Prediction/tree/main
[3] https://archive.ics.uci.edu/dataset/17/breast+cancer+wisconsin+diagnostic

4.2 Data Description

METABRIC. The METABRIC dataset comprises 33 columns and 2,509 patient records. Each row represents a clinical profile including the patient's age at diagnosis, type of breast cancer, and survival status. The features include continuous variables (age at diagnosis, tumor size, survival months), categorical variables (cancer subtype, surgery type), and boolean indicators (chemotherapy, hormone therapy). Approximately 30% missingness was identified in the 3-Gene Classifier Subtype (29.7%) and Tumor Stage (28.7%). Class imbalance was observed in vital status (800 living vs. 500 died of disease) and cancer type (>1,750 Breast Invasive Ductal Carcinoma vs. <50 Metaplastic Breast Cancer).

WDBC. The Wisconsin Diagnostic dataset comprises 569 patients and 30 numeric features plus a binary target variable describing whether the cancer is benign or malignant. Key features include cell nucleus measurements such as radius, texture, perimeter, area, and smoothness. Moderate class imbalance was observed in diagnosis (∼63% benign, ∼37% malignant).

4.3 Preprocessing Steps

4.3.1 Wisconsin Breast Cancer Dataset (Binary Classification)

To prepare for binary classification, an extraneous empty column was removed. The categorical diagnosis variable was standardized by converting to uppercase and then encoding into a binary numeric target (0 = Benign, 1 = Malignant). No missing values were identified. All 30 numeric features were standardized using z-score normalization (`StandardScaler`) within the model pipeline to ensure comparability across features with differing scales. The dataset was partitioned into training, validation, and test sets (70%/15%/15%) using stratified sampling to preserve class distribution across all splits. Class imbalance was mitigated by applying `class_weight="balanced"` to both classifiers, which inversely weights classes proportional to their frequency.

4.3.2 METABRIC Dataset (Risk Stratification and Survival Analysis)

Risk Stratification. Duplicate records were removed and rows with missing target values were excluded. To prevent data leakage, all survival-related columns were excluded from the feature set prior to model training. A synthetic three-class risk label (Low/Medium/High) was derived through quantile-based binning of Overall Survival (Months), yielding approximately 660 samples per class. Remaining missing values were imputed using column median for numeric features and most-frequent value for categorical features. An additional engineered feature, Age^2, was created to capture non-linear

age effects. Categorical variables were encoded using one-hot encoding, and the dataset was partitioned 80%/20% (train/test) with stratification on the risk label.

Survival Analysis. Numeric features—including tumor size, survival duration, and prognostic indices—were imputed using grouped mean or median values. Categorical features—including ER Status, HER2 Status, PR Status, and treatment indicators—were imputed using grouped mode. Integrative Cluster was excluded due to multicollinearity with other features. All remaining categorical variables were ordinally encoded through label encoding, and binary event indicators (Overall Survival Status, Relapse Free Status) were mapped to $\{0, 1\}$. The dataset was split 80%/20% (train/test) stratified by Cancer Type Detailed, with five-fold stratified cross-validation applied over the training partition.

5 Selection of Machine Learning Models

5.1 Overview of Models Considered

Model selection was guided by three core principles: predictive performance, interpretability for clinical use, and suitability for survival and risk analysis on structured tabular data. Deep learning architectures were reviewed but ruled out early: convolutional neural networks (CNNs) require spatial or image data, recurrent neural networks (RNNs) assume dense longitudinal sequences, and transformer-based models (BERT, GPT) are designed for unstructured text—none of which match the structured numerical and categorical nature of this dataset. Additionally, deep learning models demand far larger training sets than clinical breast cancer registries typically provide, introducing unacceptable overfitting risk. The project therefore focused entirely on traditional machine learning and statistical survival models.

5.2 Model Descriptions

Logistic Regression (Model A). Logistic Regression was selected as a baseline and co-primary classifier for binary benign/malignant classification. Its key strength is producing well-calibrated probability outputs, directly supporting the confidence-calibrated predictions the model is designed to deliver. Coefficients map directly to log-odds of malignancy, making results immediately interpretable by clinicians. Its linear decision boundary is a known limitation for complex feature interactions, but its transparency and reliability as a benchmark made it an essential inclusion.

Random Forest (Model A). Random Forest was selected as the primary ensemble classifier for Model A. By aggregating predictions across hundreds of decision trees, it

overcomes the overfitting tendency of individual trees while handling mixed data types, missing values, and outliers—all common in real-world clinical data. Feature importance scores provide a practical interpretability mechanism, identifying which clinical variables most strongly drive malignancy predictions. Pairing it with Logistic Regression enables a direct trade-off comparison between a transparent linear model and a more powerful ensemble approach.

XGBoost (Model B). XGBoost was selected for the three-class risk stratification task. Its sequential tree-building mechanism—where each tree corrects residual errors of the prior ensemble—consistently outperforms parallel ensemble methods such as Random Forest on structured tabular data, particularly where boundaries between risk tiers are subtle and nonlinear. It natively handles missing values, supports regularization to limit overfitting, and provides feature importance metrics.

Cox Proportional Hazards (Model C). The Cox Proportional Hazards (CoxPH) model was selected as the clinical gold standard for time-to-event analysis. It models the hazard function—the instantaneous risk of disease progression at any time point—while correctly incorporating censored observations, which are unavoidable when not all patients have experienced the outcome by study end. Its outputs, expressed as hazard ratios, are directly familiar and actionable for oncologists. The proportional hazards assumption was validated diagnostically to ensure model validity across patient subgroups.

5.3 Final Model Selection Summary

Table 1: Summary of selected models and their corresponding tasks.

Model	**Task**	**Algorithm(s)**
Model A	Benign vs. Malignant Classification	Logistic Regression + Random Forest
Model B	Low / Medium / High Risk Stratification	XGBoost
Model C	Disease Progression & Survival Analysis	Cox Proportional Hazards

5.4 Selection Criteria

Models were selected based on four criteria. *Clinical interpretability* was prioritized across all models—Logistic Regression and Cox produce clinically meaningful outputs directly, while Random Forest and XGBoost support interpretability through feature importance and SHAP values. *Data suitability* favored traditional machine learning methods given the structured tabular format, moderate dataset size, and mixed feature types. *Predictive*

performance was assessed via cross-validation, with the Logistic Regression vs. Random Forest pairing in Model A enabling direct benchmarking of linear against nonlinear approaches. *Censored data handling* was a non-negotiable requirement for Model C, making the Cox model the only statistically valid choice.

6 Model Development and Training

6.1 Architecture and Configuration

The project incorporates both classification models for risk stratification and a survival analysis model for time-to-event prediction, enabling a comprehensive understanding of patient outcomes.

6.1.1 Classification Models

Several supervised machine learning models were implemented, including Logistic Regression, Random Forest, and XGBoost. The feature space consisted of clinical variables such as age, tumor size, cancer stage, and treatment-related attributes, resulting in approximately 15–25 engineered features after preprocessing. Feature selection was performed using correlation analysis and model-based importance (Random Forest feature importance). Tree-based ensemble models were prioritized due to their ability to capture non-linear relationships, handle mixed data types, and provide robust performance with minimal feature scaling.

Model complexity was controlled through maximum tree depth, minimum samples per split, and regularization parameters (for XGBoost).

Model B — XGBoost Architecture. Model B was implemented as a multi-class XGBoost classifier predicting Low, Medium, or High risk tiers, derived by dividing patients into three equal quantile groups based on Overall Survival Months—with shorter survival mapped to higher risk. The target labels were numerically encoded as Low = 0, Medium = 1, High = 2 for compatibility with XGBoost's multi-class objective.

Preprocessing was handled via a `scikit-learn` `ColumnTransformer` pipeline. Numerical features were imputed using the median strategy. Categorical features were imputed using the most-frequent strategy and then one-hot encoded with unknown category handling enabled, ensuring the pipeline generalizes to unseen categories at inference time. The entire preprocessing and model pipeline was encapsulated in a single `Pipeline` object, guaranteeing that no preprocessing steps were applied to test data before being fit on training data, thereby preventing data leakage.

The resulting risk distribution across the test set was well-balanced, with 132–133 patients per class, reflecting the quantile-based label construction strategy.

6.1.2 Survival Analysis Model

The Cox Proportional Hazards model was used for survival prediction due to its interpretability and strong clinical relevance. The model assumes that covariates have a multiplicative effect on the hazard rate. Input features included the same clinical variables used in classification, ensuring consistency across models. No explicit baseline hazard function is assumed, making the model semi-parametric.

6.2 Training Process

An 80/20 train–test split was applied as the primary partition, with the 80% training portion further used in cross-validation during model selection and hyperparameter tuning. The test set was held out entirely and only used for final performance evaluation, ensuring no data leakage between development and assessment phases. For Model C, the survival dataset was split with careful attention to the distribution of censored vs. uncensored observations across partitions, ensuring each split maintained a representative proportion of events to support stable Cox model estimation.

6.3 Handling Overfitting

The following techniques were applied across models to control overfitting:

- **Cross-validation**: Stratified k-fold cross-validation was applied during training for robust, unbiased performance estimation across all folds.
- **Regularization**: L_2 regularization was applied in Logistic Regression. XGBoost incorporated both L_1 (α) and L_2 (λ) leaf weight penalties directly within its boosting objective, alongside shrinkage via a low learning rate of 0.05.
- **Tree complexity constraints**: Maximum depth was capped at 4 for XGBoost and controlled similarly in Random Forest, with minimum samples per split enforced to prevent memorization of small training subgroups.
- **Subsampling**: Row subsampling (`subsample` = 0.9) and column subsampling (`colsample_bytree` = 0.9) were applied at each XGBoost boosting round, introducing controlled stochasticity that reduces variance without significantly increasing bias.

For survival models, penalized Cox regression helps reduce overfitting in high-dimensional settings.

6.4 Transfer Learning

Transfer learning was not applied in this project, as the dataset consists of structured tabular clinical data and pre-trained models are less applicable compared to NLP or

image-based tasks. However, model generalization was enhanced through feature engineering and robust validation strategies.

6.5 Hyperparameter Tuning

To optimize model performance, hyperparameter tuning was conducted using grid search (exhaustive search over predefined parameter ranges, applied to simpler models such as Logistic Regression) and random search (efficient exploration of larger parameter spaces, used for ensemble models).

Model B — XGBoost Hyperparameters. The tuned hyperparameters for Model B are summarized in Table 2.

Table 2: XGBoost hyperparameter search space and final selected values.

Hyperparameter	Search Range	Final Value
`n_estimators`	{100, 300, 600}	600
`learning_rate`	{0.01, 0.05, 0.1, 0.3}	0.05
`max_depth`	{3, 4, 5, 7}	4
`subsample`	{0.7, 0.8, 0.9, 1.0}	0.9
`colsample_bytree`	{0.7, 0.8, 0.9, 1.0}	0.9
λ (L_2)	{1.0, 1.5, 2.0}	Tuned via CV
α (L_1)	{0, 0.1, 0.5, 1.0}	Tuned via CV

Models were selected based on validation performance, stability across folds, and interpretability. Automated machine learning (AutoML) techniques were not used, as manual tuning provided better control over model behavior and improved interpretability for clinical use.

7 Evaluation and Comparison

7.1 Evaluation Metrics

Cancer Type Classification (Model A). Model A is evaluated using Accuracy, Precision, Recall (Sensitivity), F_1-Score, ROC-AUC, and Calibration Curves. Recall is weighted most heavily, as missed malignancies carry far greater clinical harm than false positives.

Risk Level Stratification (Model B). Model B is evaluated using Macro and Weighted F_1-Score, Multi-class ROC-AUC (One-vs-Rest), and a Confusion Matrix. Weighted met-

rics account for class imbalance, while the confusion matrix reveals inter-tier misclassification patterns with different clinical severities depending on direction.

Survival Analysis (Model C). Model C is evaluated using the Concordance Index (C-statistic), Kaplan–Meier survival curves stratified by risk group, Brier Score for time-dependent calibration, and log-rank tests for statistical comparison across subgroups.

7.2 Trade-offs Across Models

The three models collectively balance accuracy, robustness, and interpretability across the care pathway. Logistic Regression prioritizes transparency and probability calibration at a modest cost to predictive ceiling. Random Forest improves performance through ensemble aggregation while retaining interpretability. XGBoost maximizes multi-class classification performance, with interpretability preserved via SHAP analysis. The Cox model trades some predictive flexibility for statistical rigor, correct censored-data handling, and clinical familiarity—trade-offs well justified where model trust and transparency are as important as raw accuracy.